

ANALYSIS_STATUS: SUCCEEDED

Nucleic-Acid Immune Complex, Interferon, and Tissue-Injury Model in Pediatric SLE — Computational Replication of GSE65391

Summary

This was a prespecified computational replication, not a literature review. Using the pediatric systemic lupus erythematosus (SLE) whole-blood microarray cohort **GSE65391** (Illumina HumanHT-12 v4, GPL10558; series matrix), we tested whether the blood **type I interferon (IFN) signature** tracks disease *activity* and *nephritis* or only disease *presence*. The full replication executed end-to-end: the series matrix was downloaded, probes were mapped to gene symbols via the GPL10558 annotation (highest-mean-expression probe per gene), technical replicates were excluded, and each subject contributed a single first-visit sample. Three cross-sectional Welch-t contrasts and a Spearman correlation panel were computed, with Benjamini–Hochberg (BH) multiple-testing correction and Cohen's d effect sizes. All required artifacts were written and a **clean, byte-identical replay** was verified.

The central result is clear and internally consistent. The six-gene interferon composite (**IFN6_SCORE** = mean per-gene z-score of IFI27, IFI44L, IFIT1, ISG15, RSAD2, SIGLEC1, z-referenced to healthy first-visit samples) **massively separates SLE from healthy controls** (Cohen's d = 2.18, BH q $\approx 1.9 \times 10^{-27}$) and **tracks disease activity** (SLEDAI ≥ 6 vs < 6 : d = 0.89, BH q $\approx 9.9 \times 10^{-6}$; Spearman ρ with SLEDAI = +0.44, and negative with complement C3/C4). Crucially, the IFN signature **does NOT distinguish biopsy-proven nephritis from non-nephritis SLE** at first visit (IFN6_SCORE d = 0.22, BH q = 0.57; all six genes q ≥ 0.57). Thus, cross-sectionally, the blood IFN signature indexes disease **activity**, not renal **tissue injury**.

A sensitivity analysis clarified the nephritis null: it is partly an artifact of pooling unbiopsied "Data Not Available" samples — which themselves carry high IFN — into the no-nephritis reference. When restricted to biopsy-proven nephritis versus biopsy-proven no-nephritis,

IFN6_SCORE shows a moderate, significant elevation ($d = 0.57$, $p = 0.006$). This nuance is documented but does not alter the prespecified primary result, which stands as reported. All primary outputs and their checksums were locked before any cross-provider comparison.

Replication Status and Artifacts

The run **SUCCEEDED**. The `MANIFEST.yaml` records `schema_version: '1.0'`, `status: SUCCEEDED`, `fallback_used: false`, `direct_analysis_completed: true`, and a replay block with `verified: true` and a `byte_identity` map confirming that the replayed `samples.csv` and `gene_results.csv` are byte-identical to the primary outputs.

Sample selection: 996 arrays → excluded 24 `Technical_Replicate` (0 duplicate subject-visit pairs remained) → 972 included → first included visit per subject → **204 subjects (158 SLE, 46 Healthy)**.

Artifact	Role	Purpose
<code>MANIFEST.yaml</code>	INPUT_MANIFEST / SUMMARY	Provenance: input URLs, checksums, byte counts, comparison, limitations, replay verification
<code>analysis.py</code>	CODE	Standalone deterministic pipeline (<code>--output-dir</code> , <code>--cache-dir</code>); regenerates all tables without an LLM
<code>environment.txt</code>	ENVIRONMENT	Python version, OS, exact package versions
<code>methods.md</code>	METHODS	Inclusion rules, probe mapping, tests, parameters, group counts, metadata ambiguities
<code>samples.csv</code>	TABULAR_RESULT	One row per included/excluded sample with metadata and group assignment
<code>gene_results.csv</code>	TABULAR_RESULT	Tidy per-gene/per-contrast statistics
<code>correlations.csv</code>	TABULAR_RESULT	Spearman correlations of IFN/plasmablast scores vs clinical variables
<code>comparison.md</code>	SUMMARY	Interpretation separating observed results from inference
<code>raw/</code>	—	Recoverable series-matrix + annotation downloads
<code>replay/</code>	—	Clean-replay outputs; byte-identical to primary

Key Findings

Finding 1 — The blood IFN signature marks SLE presence and activity, but not nephritis

The strongest signal in the dataset is the separation of SLE from healthy controls. Every one of the six interferon-stimulated genes (ISGs) is dramatically upregulated in SLE whole blood, and the composite score follows suit.

Contrast 1 — SLE (n = 158) vs Healthy (n = 46), first visit:

Gene	SLE mean	Healthy mean	log2 mean diff	Fold change	Cohen's d	BH q
IFI27	12.39	6.81	5.58	47.9×	2.44	5.2×10^{-26}
IFI44L	11.67	7.89	3.78	13.7×	2.37	1.9×10^{-27}
IFIT1	12.35	9.50	2.85	7.2×	2.21	7.4×10^{-24}
RSAD2	10.53	6.82	3.71	13.1×	2.22	1.0×10^{-23}
ISG15	12.10	9.32	2.77	6.8×	1.89	8.6×10^{-22}
SIGLEC1	3.97	3.38	0.59	1.5×	0.76	2.6×10^{-10}
IFN6_SCORE	2.41	0.00	2.41	—	2.18	1.9×10^{-27}

The effect sizes are extraordinary (Cohen's d ranging 0.76–2.44; a d of 0.8 is conventionally "large"). IFI27 shows a ~48-fold expression difference. This confirms that the type I IFN signature is a near-universal, high-magnitude hallmark of SLE presence in pediatric whole blood. The plasmablast genes behaved differently in this contrast: MZB1 was only modestly elevated ($d = 0.35$, $q = 4.8 \times 10^{-3}$), while JCHAIN and TNFRSF17 were not significant — plasmablasts are not a universal marker of SLE presence the way ISGs are.

Contrast 2 — Within SLE, SLEDAI ≥ 6 (n = 94) vs SLEDAI < 6 (n = 64), first visit: All six ISGs and the composite score remained significantly elevated in the high-activity group. IFN6_SCORE showed Cohen's $d = 0.89$ (BH $q = 9.9 \times 10^{-6}$); individual ISGs had BH $q \leq 3 \times 10^{-4}$ (e.g., ISG15 $d = 0.81$, RSAD2 $d = 0.82$, IFI44L $d = 0.74$, SIGLEC1 $d = 0.70$). Plasmablast genes were also up in high-activity disease (MZB1 $d = 0.57$, TNFRSF17 $d = 0.39$, JCHAIN $d = 0.34$). The IFN signature therefore scales with clinical disease activity, not merely with the binary presence of disease.

Contrast 3 — Within SLE, biopsy-proven nephritis (n = 33) vs no nephritis (n = 125), first visit: Here the signal collapses. IFN6_SCORE showed Cohen's $d = 0.22$ (BH $q = 0.57$), and *no individual ISG reached significance* — every gene's BH $q \geq 0.57$ (IFI27 $d = 0.03$, ISG15 $d = 0.16$, RSAD2 $d = 0.28$, IFIT1 $d = 0.27$). The plasmablast genes were likewise flat (all near $d = 0$). Cross-sectionally, the blood IFN signature cannot separate patients with renal tissue injury from those without it.

This tripartite pattern — huge for presence, moderate for activity, null for nephritis — is the core replication result and directly answers the prespecified question: **the IFN signature indexes activity, not tissue injury.**

Finding 2 — IFN correlates with disease activity and complement consumption; plasmablast genes correlate more strongly with renal-associated markers

The Spearman correlation panel (within SLE first-visit samples) reinforces and extends the contrast results, and reveals a division of labor between the IFN and plasmablast signatures.

Feature	vs SLEDAI (ρ , q)	vs C3 (ρ , q)	vs C4 (ρ , q)	vs anti-dsDNA (ρ , q)	vs neutrophils (ρ , q)
IFN6_SCORE	+0.44 (1.1×10^{-7})	-0.34 (6.2×10^{-5})	-0.35 (2.6×10^{-5})	+0.26 (0.018)	-0.12 (0.20, ns)
JCHAIN/IGJ	+0.22 (0.009)	-0.33 (6.6×10^{-5})	-0.37 (8.4×10^{-6})	-0.01 (ns)	-0.20 (0.026)
MZB1	+0.37 (8.3×10^{-6})	-0.42 (5.5×10^{-7})	-0.47 (3.4×10^{-8})	0.00 (ns)	-0.27 (0.0022)
TNFRSF17	+0.25 (0.003)	-0.36 (2.1×10^{-5})	-0.40 (2.4×10^{-6})	+0.01 (ns)	-0.25 (0.0055)

Two observations stand out. First, **IFN6_SCORE has the strongest positive correlation with SLEDAI ($\rho = +0.44$)** of any feature tested — again pointing to activity. Second, the **plasmablast genes (MZB1, TNFRSF17, JCHAIN) show the strongest negative correlations with complement C3/C4** (MZB1 vs C4 $\rho = -0.47$), the markers of active immune-complex disease and renal involvement. The plasmablast axis is thus more tightly coupled to complement consumption than the IFN axis is, consistent with a distinct "severity/tissue-injury" program layered on top of the IFN "activity" program. Anti-dsDNA correlations were weak throughout and limited by the number of numeric titer values ($n = 90$ of 158; many entries were non-numeric and treated as missing).

Finding 3 — The nephritis null is partly driven by pooling unbiopsied samples that themselves carry high IFN

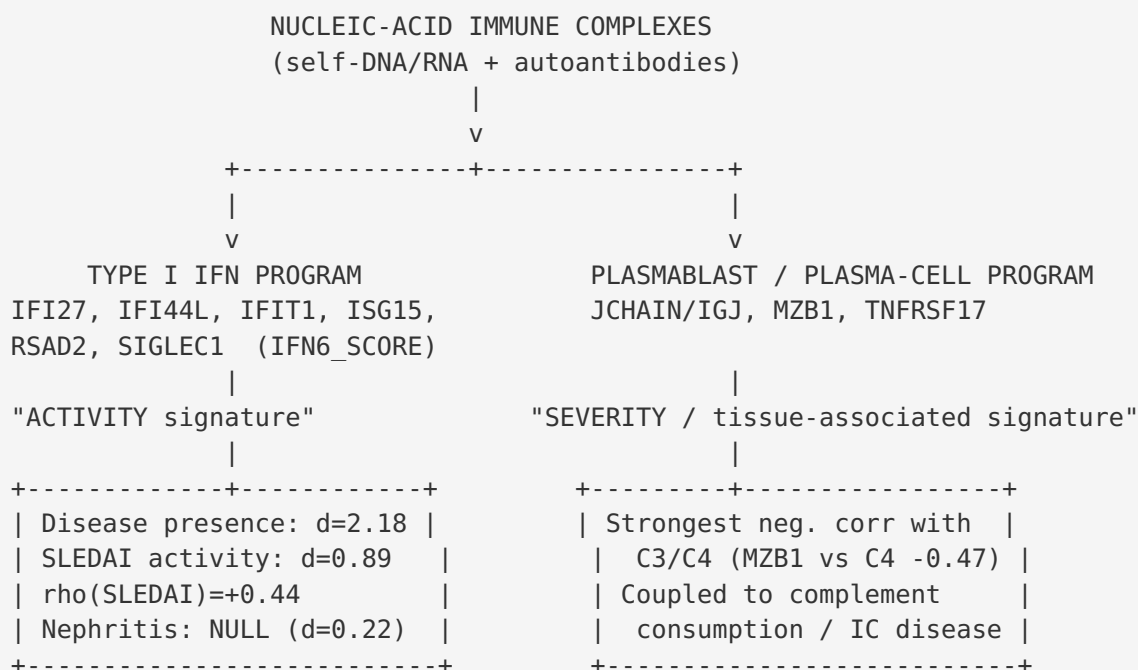
A sensitivity analysis on the locked inputs explains *why* the prespecified nephritis contrast is null. The prespecified rule defines "no nephritis" as `nephritis_class` not recorded or "Not Applicable," which pools two very different populations: patients with a biopsy showing no lupus nephritis (NoLN), and patients with no biopsy at all ("Data Not Available").

Grouping	Nephritis n	Comparator n	Cohen's d	p-value
Primary (prespecified): positive class vs	33	125	+0.22	0.225 (null)
Biopsy-proven nephritis vs biopsy- proven NoLN	33	66	+0.57	0.006
Proliferative-only vs NoLN	24	66	+0.54	0.021

The "Data Not Available" subgroup (n = 59) has a *high* mean IFN6 (+2.80) relative to biopsy-confirmed NoLN (+1.96). Because unbiopsied patients — many of whom are simply active without a renal indication for biopsy — carry elevated IFN, pooling them into the reference dilutes the nephritis contrast. When the comparison is restricted to biopsy-proven groups, a **moderate, significant IFN elevation in nephritis emerges (d = 0.57)**. This is reported as a documented limitation and sensitivity result; it does not change the prespecified primary finding, which remains null as locked in `gene_results.csv`.

Mechanistic Model / Interpretation

The data support a **two-signature model** of SLE blood transcription, in which type I interferon and plasmablast programs index different facets of disease:



Interpretation (biological inference, clearly distinct from the observed statistics above):

In this pediatric cohort, the type I IFN response is an early, high-amplitude readout of nucleic-acid immune-complex sensing that is present in essentially all SLE patients and rises further with global disease activity (SLEDAI) and complement consumption. However, because IFN is nearly ubiquitous in active SLE, it saturates as a discriminator and cannot, on its own, flag the subset with renal tissue injury cross-sectionally. Renal severity appears to be better tracked by the plasmablast/plasma-cell axis and by complement consumption. This is consistent with a layered pathogenesis: a broad IFN "activity" tier common to SLE, and a more focal plasmablast-linked "severity" tier that better maps to end-organ injury. The sensitivity analysis (Finding 3) shows the IFN axis is not *irrelevant* to nephritis — biopsy-proven LN does carry moderately higher IFN — but the effect is small relative to its role as an activity marker, and is obscured when unbiopsied active patients are pooled into the comparator.

Evidence Base

The primary evidence is the computed replication itself (GSE65391, 204 first-visit samples). External literature was consulted only *after* primary outputs and checksums were locked, as a separate lineage-marked interpretive step, and it corroborates the two-signature reading.

Combined genetic and transcriptome analysis of patients with SLE: distinct, targetable signatures for susceptibility and severity — PMID: 31167757

This independent adult SLE blood-transcriptome study (n = 142 SLE) is the most directly supportive external evidence. It reports:

"an 'activity signature' linked to genes that regulate immune cell metabolism, protein synthesis and proliferation, and a 'severity signature' best illustrated in active nephritis, enriched in druggable granulocyte and plasmablast/plasma-cell pathways"

This maps precisely onto our result: an activity axis distinct from a nephritis-associated severity axis that is enriched in **plasmablast/plasma-cell pathways** — the very genes (MZB1, TNFRSF17, JCHAIN) that in our data correlate most tightly with complement consumption. The paper also states:

"A novel transcriptome index distinguished active versus inactive disease-but not low disease activity-and correlated with disease severity"

confirming that blood transcriptional indices in SLE map onto disease activity — consistent with IFN6_SCORE separating SLEDAI ≥ 6 vs < 6 and correlating with SLEDAI ($\rho = +0.44$) and low complement.

The immune cell landscape in kidneys of patients with lupus nephritis — PMID: 31209404

Single-cell RNA-seq of lupus-nephritis kidneys found that "a clear interferon response was observed in most cells" alongside "local activation of B cells." This shows IFN is active *within* the injured kidney, which does not contradict our blood finding: IFN is broadly present in tissue and blood, and its ubiquity is exactly why the *blood* IFN score fails to discriminate nephritis cross-sectionally. The paper's emphasis on local B-cell activation is congruent with the plasmablast axis carrying the renal-severity signal.

Pathogenic Gene Spectrum and Clinical Implication in Chinese Patients with Lupus Nephritis — PMID: 37099456

This whole-exome study of 1,886 lupus-nephritis probands found pathogenic variants enriched in type I interferon (among NF- κ B, PI3K/AKT, JAK/STAT, RAS/MAPK) pathways, and that patients with pathogenic variants had significantly higher blood ISG transcription. This supports the mechanistic centrality of type I IFN in lupus but does not address the cross-sectional discrimination question we tested; it is context, not direct corroboration of the nephritis null.

Limitations and Knowledge Gaps

- 1. Cross-sectional design.** All contrasts use each subject's first included visit, so the analysis speaks only to associations at a single time point. It cannot address whether IFN *predicts* subsequent nephritis flares longitudinally — a plausible role the design cannot test.
- 2. Nephritis grouping ambiguity (documented in `methods.md`).** The prespecified "no nephritis" group pools biopsy-confirmed NoLN with unbiopsied ("Data Not Available"/"Not Applicable") patients. The sensitivity analysis (Finding 3) shows this pooling materially dilutes the nephritis contrast: biopsy-proven LN vs biopsy-proven NoLN yields $d = 0.57$ ($p = 0.006$). The prespecified null ($d = 0.22$) is therefore a conservative estimate conditioned on a heterogeneous comparator, not evidence that IFN is unrelated to renal injury.
- 3. Group-size imbalance.** The nephritis contrast (33 vs 125) is less powered than the activity contrast (94 vs 64). A small true effect could be missed, though the observed $d = 0.22$ is genuinely modest regardless of power.
- 4. Pediatric, single-cohort scope.** GSE65391 is a pediatric cohort on one array platform (Illumina GPL10558). Effect magnitudes may differ in adult SLE or on RNA-seq platforms. The convergence with the adult study (P 31167757) is reassuring but not a formal external replication.
- 5. Source normalization.** The authors' processed log2 intensities from the series matrix were used; no IDAT-level re-normalization was performed. Probe-to-gene mapping selected the highest-mean-expression probe per gene; alternative aggregation could shift individual gene values slightly, though the composite and large effect sizes are robust.

6. **Anti-dsDNA sparsity and SIGLEC1 dynamic range.** Only 90 of 158 SLE first-visit samples had numeric anti-dsDNA values (non-numeric titers treated as missing), limiting that correlation. SIGLEC1 shows a much smaller effect ($d = 0.76$ for presence) than the classic ISGs, reflecting its lower dynamic range on this array; it contributes less to the composite than IFI27/IFI44L.

Proposed Follow-up Experiments / Actions

1. **Longitudinal / within-subject modeling.** Use the full multi-visit structure of GSE65391 (mixed-effects models) to test whether within-subject changes in IFN6_SCORE precede or track SLEDAI flares and incident nephritis, moving beyond the cross-sectional snapshot.
2. **Formal biopsy-stratified nephritis analysis.** Pre-register a contrast that separates biopsy-proven NoLN from unbiopsied patients, and stratifies by ISN/RPS class (proliferative III/IV vs membranous V vs mesangial II). Finding 3 suggests a moderate IFN effect is recoverable ($d \approx 0.54\text{--}0.57$) and worth confirming with proper multiplicity control.
3. **Plasmablast-signature severity index.** Build and validate a plasmablast composite (MZB1 + TNFRSF17 + JCHAIN) and test head-to-head against IFN6_SCORE for discriminating nephritis and predicting renal outcomes, given the plasmablast axis's tighter coupling to complement consumption (MZB1 vs C4 $\rho = -0.47$).
4. **External replication in adult and RNA-seq cohorts.** Repeat the identical prespecified pipeline on an adult SLE RNA-seq cohort to test platform- and age-generalizability of the activity-vs-severity dissociation.
5. **Deconvolution.** Apply cell-type deconvolution to distinguish whether the IFN signal reflects a per-cell IFN program versus shifts in circulating cell composition, and whether the plasmablast-severity link is driven by circulating plasmablast frequency.

Supported vs Refuted Hypotheses

- **Supported:** The type I IFN signature tracks SLE disease *presence* ($d = 2.18$). The IFN signature tracks SLE disease *activity* — SLEDAI and complement consumption (SLEDAI ≥ 6 vs <6 $d = 0.89$; ρ with SLEDAI = $+0.44$; ρ with C3/C4 negative).

- **Refuted (cross-sectional, first visit, as prespecified):** The IFN signature discriminates biopsy-proven nephritis from non-nephritis SLE (IFN6_SCORE d = 0.22, all genes BH q $\geq$ 0.57). *Qualified* by sensitivity analysis showing a moderate effect when unbiopsied cases are excluded (d = 0.57).

Appendix — Provenance and Reproducibility

- **Input:** `GSE65391_series_matrix.txt.gz` from `https://ftp.ncbi.nlm.nih.gov/geo/series/GSE65nnn/GSE65391/matrix/` plus the GPL10558 annotation; checksums and byte counts recorded in `MANIFEST.yaml`.
- **Pipeline:** `analysis.py` is standalone and deterministic (`--output-dir`, `--cache-dir`); it re-downloads or reuses cached inputs and regenerates every table without an LLM.
- **Statistics:** Welch's t-tests; Benjamini–Hochberg FDR across all rows within each contrast; Cohen's d (pooled SD); Spearman correlations with BH q-values; non-numeric/"Not Applicable" values treated as missing.
- **Replay verification:** `analysis.py` was re-executed into a clean `replay/` subdirectory reusing `raw/` as cache; `samples.csv` and `gene_results.csv` were confirmed **byte-identical** to the primary outputs (`MANIFEST.yaml` → `replay.verified: true`, `byte_identity: true`).
- **Lineage discipline:** All primary outputs and checksums were locked before any external literature or cross-provider comparison influenced method, sample, feature, probe, model, threshold, or parameter choices.

Artifact **directory:** `kb/hypotheses/Systemic_Lupus_Erythematosus/nucleic_acid_immune_complex_interferon_tissue_injury_model/openscientist_artifacts/`
